

DATABASE MANAGEMENT SYSTEMS (CS301)

Nested Queries

Learning Objectives

Understand subqueries, nested queries, correlated queries, and SQL operators used within subqueries.

Introduction to Nested Queries

A nested query (subquery) is a query written inside another SQL query. The inner query executes first and its result is used by the outer query.

Subqueries

Subqueries can be used in SELECT, INSERT, UPDATE, and DELETE statements to retrieve intermediate results.

Single Row Subquery

Returns only one row and is commonly used with operators such as =, >, <, >=, and <=.

Example: `SELECT Name FROM Student WHERE Marks > (SELECT AVG(Marks) FROM Student);`

Multiple Row Subquery

Returns multiple rows and is commonly used with IN, ANY, and ALL operators.

IN Operator

Checks whether a value exists within a list returned by a subquery.

`SELECT Name FROM Student WHERE DeptID IN (SELECT DeptID FROM Department);`

EXISTS Operator

Returns TRUE if the subquery returns one or more rows.

NOT EXISTS Operator

Returns TRUE when the subquery returns no rows.

ANY Operator

Compares a value with any value returned by the subquery.

ALL Operator

Compares a value with all values returned by the subquery.

Correlated Subqueries

A correlated subquery depends on values from the outer query and executes once for each row processed by the outer query.

Applications

- Data Filtering
- Report Generation
- Performance Analysis
- Business Intelligence Queries

Exam Important Questions

1. Define nested query.
2. Explain single-row and multiple-row subqueries.
3. Differentiate IN and EXISTS.
4. Explain correlated subqueries.
5. Write SQL queries using ANY and ALL.

Summary

Nested queries simplify complex data retrieval tasks by allowing one query to use the results of another query. They are widely used in SQL-based database applications.